FITSai

FRIENDLY INTELLIGENT TECHNICAL SUPPORT AI

Your company's AI-powered assistant with gamified progression, voice interaction, and intelligent document search.

50	30	31	Smart
Levels	Personalities	Achievements	AI Routing

Complete Feature Guide • February 2026 • Internal Use

Codename: FELICIA (Fully Engineered Linguistic Intelligence and Conversational Assistant)

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1. Chat & AI Intelligence

The core of FITSai - an intelligent assistant that learns and adapts

Smart AI Routing User

FITSai uses a smart routing system that automatically selects the best AI model for each question, optimising for speed, cost, and quality.

Query Type	Model	Where	Speed
Simple questions	llama3.1:8b via Ollama	Local GPU	1-3s
Coding questions	deepseek-coder-v2 via Ollama	Local GPU	2-5s
Knowledge base answers	llama3.1:8b (local first, GPT-4o-mini fallback)	Local / Cloud	1-4s
Complex reasoning	GPT-4o-mini (OpenAI)	Cloud	1-3s
Image analysis	GPT-4o-mini (OpenAI)	Cloud	2-4s

How it works: When you ask a question, FITSai analyses the complexity (word count, keywords, context) and routes it to the most appropriate model. Simple questions run on the local GPU for free, while complex reasoning and image analysis use cloud AI. If the local model is unavailable, everything gracefully falls back to OpenAI.

Knowledge Base (RAG) User

FITSai searches your company's uploaded documents before answering, so responses are grounded in real company knowledge.

- Upload PDF, DOCX, TXT, or image files (up to 50MB)
- Documents are automatically chunked and indexed for semantic search
- Answers include confidence scores and source references
- Previously approved answers are cached to save costs

Cost Savings: When you give a thumbs-up to an answer, it's saved in the learned answer database. Similar future questions reuse this answer at zero API cost. Local model routing further reduces costs by handling simple queries on-device.

Self-Correction & Quality Grading User

Every response from a local AI model is automatically quality-checked in the background by GPT-4o-mini:

- Local model answers are graded for accuracy and completeness
- Poor-quality answers are corrected and saved to the knowledge base
- Future similar questions retrieve the corrected answer via RAG
- Thumbs-down feedback triggers immediate re-grading (bypasses sampling)

This means local model answers are never unchecked — they go through the same quality pipeline as cloud answers, just asynchronously.

Streaming Responses User

Responses appear in real-time as they're generated, token by token - no waiting for the full answer. Uses Server-Sent Events (SSE) for smooth, live text delivery. Full conversation thread context is maintained so FITSai understands your entire discussion.

Conversation Threading & Sharing User

Organise your conversations into separate threads by topic. FITSai remembers context within each thread, so you can switch between projects without losing your place.

- Create new threads or continue existing ones
- Search across all your conversation history
- Soft-delete conversations (recoverable, no data loss)
- Export chats as PDF, JSON, or TXT
- **Share conversations** — copy a formatted thread to clipboard for sharing with colleagues

Chat Templates User

Pre-defined prompts for common tasks, accessible via **Ctrl+T**. Templates help you get consistent, well-structured responses for recurring workflows like daily standups, code reviews, or ticket summaries.

Feedback Learning Loop User

Every response includes thumbs-up/thumbs-down buttons. Your feedback directly improves FITSai:

- **Thumbs Up:** Answer is saved for future reuse (+10 XP for you)
- **Thumbs Down:** Answer is flagged for improvement
- High-confidence matches (85%+) reuse approved answers instantly

2. Smart Colleague Intelligence

FITSai learns how you work, detects frustration, and connects you with team knowledge

Morning Briefing User

Start your day with a personalised briefing card that appears when you first open FITSai:

- Personalised greeting based on time of day and your streak
- Pending reminders due today
- Recent achievements and XP progress
- Follow-up on unresolved issues from the past 7 days
- Daily tip tailored to your usage patterns

Frustration Detection User

FITSai detects when you're getting frustrated and automatically adjusts its tone:

- Analyses keywords, CAPS LOCK, punctuation patterns, and repeated thumbs-down feedback
- Three frustration levels: mild, moderate, high
- Empathy is injected into AI context, adapted to your active personality mode
- High frustration triggers a gentler, more supportive tone without being patronising

Communication Style Adaptation User

FITSai learns your communication preferences from your feedback patterns:

- **Brevity:** Learns whether you prefer short or detailed answers
- **Formality:** Adapts between casual and formal tone
- **Technical depth:** Adjusts jargon level based on your expertise

Preferences are learned automatically from thumbs-up/down feedback — no configuration needed.

Peer Knowledge Linking User

When you ask a question, FITSai checks if a colleague in your department has solved a similar problem before. If so, it mentions who solved it and what the solution was, connecting team knowledge across conversations.

Proactive Nudges User

FITSai detects when you've asked about the same topic 3+ times and proactively suggests creating a cheat sheet or document from your repeated questions — turning tribal knowledge into permanent documentation.

3. Personality System

30 unique AI voices that transform how FITSai communicates with you

FITSai has **30 personality modes** that change its communication style while maintaining technical accuracy. New personalities unlock as you level up, giving you more options to find the voice that works best for you.

How to switch: Say "Switch to [mode name] mode" in chat, or select from your profile settings. Example: "Switch to sarcastic sidekick mode"

Personality	Style	Unlock Level
Professional Maven	Polished, efficient, corporate-friendly	Level 1
Sarcastic Sidekick	Witty, dry humor, playfully cynical	Level 3
Enthusiastic Cheerleader	High energy, motivational	Level 5
Zen Minimalist	Calm, concise, mindful	Level 7
Nerdy Explainer	Loves details, references, tangents	Level 9
Southern Charm	Warm, hospitable, colloquialisms	Level 11
Brutally Honest	Direct, no-nonsense	Level 13
Quirky Creative	Imaginative, metaphors	Level 15
Empathetic Counselor	Understanding, validating	Level 17
Mysterious Sage	Cryptic wisdom, riddles	Level 19
Chaotic Gremlin	Playful chaos, unpredictable	Level 21
Academic Scholar	Formal, well-researched	Level 23
Cool Aunt	Relaxed, supportive	Level 25
Drill Sergeant	Tough love, accountability	Level 27
Hopeless Romantic	Poetic, dreamy	Level 29
Tech Bro	Startup culture, business jargon	Level 31
Grandmother Wisdom	Comforting, traditional advice	Level 33
Conspiracy Theorist	Questions everything (still helpful)	Level 35
Game Show Host	Upbeat, engaging	Level 37
Noir Detective	Dramatic, 1940s private eye	Level 39
Bubbly Optimist	Silver linings, positivity	Level 41
Stoic Philosopher	Balanced, rational	Level 43
Mad Scientist	Experimentally-minded, unhinged	Level 44
Diplomatic Mediator	Balanced, conflict-averse	Level 45
Sassy Best Friend	Keeps it real, teases lovingly	Level 46
Vintage Librarian	Knowledgeable, organized	Level 47
Adventurous Explorer	Bold, curious	Level 48
Punk Rebel	Anti-establishment, edgy	Level 49
Mindful Yogi	Present-focused, holistic	Level 50
Existential Overthinker	Deep questions, philosophical	Level 50

4. Gamification & Progression

Level up, earn achievements, and compete on the leaderboard

50-Level Character Progression

Every interaction with FITSai earns you XP. As you level up, you unlock new personality modes, avatar colours, and earn prestigious titles.

XP Rewards

Action	XP Earned
Ask a question	+5 XP
Give feedback (thumbs up/down)	+10 XP
Upload a document	+25 XP
Project contribution	+15 XP
Unlock an achievement	+50 to +200 XP (auto-awarded)

Character Classes

Levels 1-5: Kobold (Beginner)

Levels 6-10: Goblin

Levels 11-15: Troll

Levels 16-20: Dwarf

Levels 21-25: Elf

Levels 26-30: Wizard

Levels 31-35: Phoenix

Levels 36-40: Unicorn

Levels 41-45: Dragon

⚡**Levels 46-50:**Demigod / God

Achievements & Titles

Unlock achievements by reaching milestones. Each achievement grants bonus XP and some unlock equippable titles that display on the leaderboard.

Rarity	Examples	XP Reward
Common (8)	Hello World, First Steps, Curious Mind, Dedicated	50 XP
Rare (11)	Early Adopter, Knowledge Builder, Committed	100 XP
Epic (8)	Bug Hunter, Department Champion, Elite Status	150 XP
Legendary (2)	Legendary (Level 40), The Answer	200 XP
Mythic (2)	Company MVP, Godhood (Level 50)	200 XP

Leaderboards

See how you rank against colleagues across multiple views:

- **All-Time:** Ranked by level and total XP

- **Weekly:** XP earned in the last 7 days
- **Monthly:** XP earned in the last 30 days
- **Department:** Filtered by your department
- **Most Improved:** Biggest XP gain this week
- **Department Champions:** Top user per department

Leaderboard entries show your avatar, level, character class, equipped title, and department badge.

Avatar Customisation

Personalise your avatar with colour schemes that unlock as you level up:

Level	Colour	Level	Colour
1	Default (class colour)	30	Arctic White
5	Ocean Blue	35	Shadow Black
10	Forest Green	40	Golden Glory
15	Sunset Orange	45	Rainbow Prism
20	Royal Purple	50	Cosmic Void
25	Crimson Red		

Visual Effects by Level: Level 10+ gets a pulsing glow, Level 20+ gets an animated border, Level 30+ gets particle trails, and Level 50 gets cosmic multi-layer effects.

5. Voice & Avatar

Speak to FITSai and watch the holographic avatar respond

Voice Chat User

Click the avatar or voice button to start a real-time voice conversation with FITSai.

- Speech-to-text via OpenAI Whisper (multi-language support)
- Text-to-speech with 6 voice options: Alloy, Echo, Fable, Onyx, Nova, Shimmer
- Full conversation context maintained during voice sessions
- Works on both desktop and mobile browsers

3D Holographic Avatar User

FITSai appears as an interactive 3D holographic face powered by Three.js with real-time visual feedback:

Avatar State	Visual Effect
Idle	Gentle breathing animation, particles drift slowly
Listening	Particles flow inward, glow increases, face attentive
Speaking	Particles burst outward, maximum glow, energetic
Processing	Particles swirl rapidly, contemplative state
Error	Red tint, chaotic particle movement

The avatar follows your mouse cursor, rotating up to 12 degrees to "look at" you. Shader effects include fresnel glow, scan lines, and personality-based colour tinting.

6. Productivity Tools

Reminders, documents, memory, and search to keep you productive

Reminder System User

Set reminders using natural language and receive popup notifications when they're due.

- **Create:** "Remind me to call the client at 3pm" - FITSai parses the time and title
- **Popup:** Orange notification appears in top-right corner when due
- **Actions:** Mark as Done, Snooze 15 minutes, or Snooze 1 hour
- **Overdue:** All overdue reminders keep showing until you dismiss them
- **Recurrence:** Set daily, weekly, or monthly recurring reminders

Reminders are checked every 30 seconds in the background, so you'll never miss one while using FITSai.

Document Management User

Upload and manage company documents to build the AI knowledge base.

- **Supported formats:** PDF, DOCX, TXT, PNG, JPG (up to 50MB)
- Documents are automatically chunked, embedded, and indexed
- FITSai searches your documents when answering questions
- **Document Analysis:** Summarise, analyse, extract key info, or memorise documents
- **Document Conversion:** Convert DOCX/XLSX to PDF
- Earn 25 XP for each document uploaded

Memory System User

FITSai remembers important information about you across sessions.

- Automatically extracts key facts from conversations (40+ recognition patterns)
- Memories have importance scores that decay over 30 days
- Semantic search finds relevant memories even with different wording
- View, search, and manage your memories from the Memory Manager page
- All AI responses are informed by your stored memories for personalised answers

Chat Search & Export User

Search across your entire conversation history with keyword matching and highlighting. Export any conversation as PDF, JSON, or TXT for record-keeping or sharing.

7. Admin Features (God Mode)

System administration dashboard for admins and Level 46+ users

Access Required: God Mode is available to users with the **admin** role or users who have reached **Level 46+** (Demigod class).

User Management Admin

Full control over user accounts from the God Mode dashboard:

- Search and view all users with their levels, XP, and departments
- Edit user details (name, email, department, role)
- Create new user accounts
- Award XP to users with reason tracking
- Modify character stats (STR/DEX/CON/INT/WIS/CHA)

System Tools Admin

Direct system access for troubleshooting and maintenance:

- Execute bash commands on the server
- Create and edit files
- Monitor and restart backend services
- Send test notifications
- View system-wide statistics and trends

Roadmap Management Admin

Interactive development roadmap visible in God Mode. Admins can update feature statuses (Pending / In Progress / Complete) directly from the UI. Changes are reflected for all users viewing the roadmap.

Achievement Management Admin

Search for any user and manually award or revoke achievements. Useful for recognising special contributions or correcting missed auto-awards. XP and level/class are automatically updated when achievements are awarded.

Usage Analytics Dashboard Admin

Comprehensive analytics on the Budget page:

- **Model breakdown:** Queries per model (local vs cloud), cost per model
- **Source breakdown:** RAG, local AI, OpenAI, cached answers
- **Top users:** Most active users by conversation count
- **Feedback stats:** Thumbs up/down ratios, learned answer savings
- **Department activity:** Usage breakdown by department

Code Health Monitoring Admin

Autonomous code scanning that monitors repository health. View scan results and notifications from the God Mode panel.

8. FITSai CLI

Command-line interface for developers and power users

FITSai CLI is an interactive command-line tool (similar to Claude Code) that brings FITSai's AI capabilities to your terminal.

CLI Commands

Command	Description
`fitsai`	Start interactive REPL session
`fitsai login`	Authenticate with your FITSai account
`fitsai logout`	Clear stored credentials
`fitsai chat "question"`	Quick one-off query
`fitsai config --show`	View current settings
`fitsai sessions --list`	List saved sessions
`fitsai sessions --resume ID`	Resume a previous session

CLI Tools (6 Built-in)

When FITSai needs to interact with your local system, it can use these tools with your confirmation:

Tool	Description
`read_file`	Read file contents from your project
`edit_file`	Make targeted edits to files
`write_file`	Create new files
`bash`	Execute shell commands
`glob`	Find files by pattern matching
`grep`	Search file contents with regex

All tool executions require user confirmation before running.

9. Technical Architecture

The technology stack powering FITSai

Frontend

- React 19 + TypeScript
- Vite (build tool)
- TailwindCSS (styling)
- Three.js (3D avatar)
- TanStack Query (state)
- React Router (navigation)

Backend

- FastAPI (Python)
- SQLAlchemy (async ORM)
- PostgreSQL 16 (database)
- ChromaDB (vector DB)
- Alembic (migrations)
- Ollama (local AI serving)
- OpenAI API (cloud AI)

AI Models

- llama3.1:8b via Ollama (local chat)
- deepseek-coder-v2 via Ollama (local code)
- GPT-4o-mini (cloud reasoning)
- OpenAI Whisper (voice input)
- OpenAI TTS (voice output)
- Sentence Transformers (embeddings)

Infrastructure

- Ubuntu 24.04 LTS
- Ollama (local AI serving)
- Nginx (reverse proxy)
- Docker (databases)
- systemd (services)
- NVIDIA CUDA (GPU)

Server Specifications

Component	Specification
CPU	AMD Ryzen 7 5700 (16 cores)
RAM	32 GB DDR4
GPU	NVIDIA RTX 4060 (8 GB VRAM)
OS	Ubuntu 24.04 LTS

Services

Service	Port	Managed By
Backend API	8000	systemd: felicia-backend
Ollama (AI Models)	11434	systemd: ollama
PostgreSQL Database	5432	Docker: felicia-postgres
Nginx (Web Server)	80 / 443	systemd: nginx

10. Security

Enterprise-grade authentication and data protection

Authentication

- **JWT Tokens:** HS256 algorithm, 7-day expiration
- **Password Hashing:** Bcrypt with salt rounds
- **Role-Based Access:** User, Manager, Admin roles
- **Level-Based Elevation:** Level 46+ gains God Mode access
- **Password Reset:** Email token-based forgot password flow

Data Protection

- SQL injection prevention via ORM parameterised queries
- Pydantic schema validation on all API inputs
- CORS configuration for allowed origins
- WebSocket token validation for voice sessions
- Document storage outside web root
- Soft-delete for conversations (no permanent data loss)
- Voice data not permanently stored on server

11. Quick Start Guide

Get started with FITSai in minutes

For Users

1. **Login** - Use your company email and password
2. **Ask a question** - Type in the chat box and hit Enter (+5 XP)
3. **Give feedback** - Click thumbs up or down on answers (+10 XP)
4. **Upload documents** - Add company docs to improve answers (+25 XP)
5. **Check your progress** - View level, XP, and achievements on your profile
6. **Set reminders** - Ask FITSai "Remind me to..." for natural language reminders
7. **Try voice** - Click the avatar to start a voice conversation
8. **Switch personalities** - Say "Switch to [mode] mode" for a different vibe
9. **Compete** - Check the leaderboard to see your ranking

For Admins

1. **Access God Mode** - Click the God Mode icon in the navigation (admin role required)
2. **Manage users** - Search, edit, and create user accounts
3. **Award XP** - Recognise users with XP bonuses
4. **Monitor system** - Check system stats, service health, and API costs
5. **Update roadmap** - Change feature statuses from the roadmap panel
6. **View analytics** - Check model usage, costs, and feedback stats on the Budget page

Tips for Best Results

- **Be specific:** More details = better answers
- **Give feedback:** Thumbs up/down improves FITSai and earns you XP
- **Upload documents:** The more company knowledge, the better the answers
- **Try different personalities:** Find the communication style that works for you
- **Use templates:** Press Ctrl+T for pre-built prompts
- **Ask follow-ups:** FITSai remembers your conversation context

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